

TECHNICAL FRAMEWORK — PRACTICE-LED RESEARCH

Misattribution Machine

An AI-driven tool for detecting and visualising misattribution
in digital archive systems — prototype for museum exhibition

RESEARCHER Teona Yamanidze
FOCUS Malevich · Exter · Pagava · Kakabadze · Parajanov
DATASETS 659 records across 2 datasets
DATE May 2026
REPOSITORY github.com/yamanidzet/Teona-Yamanidze

What the archive forgot

This document describes a prototype AI tool built to detect, demonstrate, and visualise misattribution in digital archive systems. The tool operates across two registers simultaneously: as a **misattribution detection engine** that queries multiple large language models (LLMs) and compares their responses against a researcher-built Record of Truth; and as an **exhibition system** designed for deployment across 12 LED screens in a museum space.

The core argument the tool demonstrates is structural: misattribution is not simply a cataloguing error made by individual institutions. It is a condition that propagates. An absence of a nationality qualifier in a Library of Congress authority record flows downstream into WorldCat, into national library aggregators such as Trove, into museum catalogues, and finally into the training data of LLMs — where it is replicated as if it were fact. Correcting the record upstream does not automatically correct what has been absorbed downstream.

The exhibition makes this process visible — without text, through distortion and clarity alone.

659 TOTAL RECORDS ANALYSED across both datasets	~94% RECORDS WITH MISATTRIBUTION wrong label, omission, or spelling	3 PRIMARY ERROR TYPES wrong label · omission · spelling
--	--	--

Dataset 1 — Parajanov Bibliographic Dataset

305 bibliographic records drawn from Trove (NLA), WorldCat/OCLC, ACMI, NGV Library, QAGOMA Cinémathèque, and State Library Victoria. Each record documents how Sergei Parajanov is identified in terms of nationality, name form, and subject headings.

Archive	System type	Misattribution found
Trove (NLA)	National aggregator	Labelled "Soviet" — Armenian identity erased
WorldCat / OCLC	Global bibliographic	Multiple transliteration variants; "Soviet" throughout
ACMI	Film institution	Labelled "Soviet" — no Armenian or Georgian attribution
NGV Library	Museum catalogue	One record (COBISS/Slovenian) correctly uses "Armenian" — the only correct
QAGOMA	Curatorial	Labelled "Soviet" — no Armenian attribution

The Record of Truth establishes the correct form for all 305 records: *Parajanov, Sarkis Hovsepi [Serge■ Iosifovich Paradzhanov, 1924–1990]*, with nationality *Soviet-Armenian (Georgian-born Armenian)*.

Dataset 2 — Malevich / Exter / Pagava / Kakabadze Artwork Dataset

354 artwork catalogue records drawn from MoMA (New York), NGA (Canberra), NGV (Melbourne), Centre Pompidou (Paris), Yale University Art Gallery, Stedelijk Museum (Amsterdam), and others.

KAZIMIR MALEVICH Labelled: <i>Russian / Soviet</i> Correct: Ukrainian-born (Kyiv); Polish ethnic heritage; b. 1879 NGA repeats wrong birth year (1878) across 200+ records. Correct year: 1879. Name inconsistency: "Kazimir" vs "Kasimir" within the same institution.	272 records
ALEXANDRA EXTER Labelled: <i>Russian</i> Correct: Ukrainian-born (Kyiv); trained Kyiv School of Art; émigré to Paris 1924 MoMA frames her work within "A Revolutionary Impulse: The Rise of the Russian Avant-Garde" — misattribution embedded in the exhibition title, not just the catalogue field.	44 records
VERA PAGAVA Labelled: <i>Empire Russe</i> Correct: Georgian-born (Tbilisi); émigré to Paris 1923; never naturalised French All 19 Centre Pompidou records use "Empire Russe" — technically a birthplace fact but functioning as imperial erasure of Georgian national identity.	19 records
DAVID KAKABADZE Labelled: <i>Absorbed into Société Anonyme "Russian avant-garde" canon</i> Correct: Georgian-born (Kutaisi/Tbilisi); studied Paris 1919–1927; returned Soviet Georgia 1927 Yale field records correctly note "Georgian" — but structural misattribution via the 1984 Herbert Catalogue Raisonné, which frames the Société Anonyme collection within a Russian/Soviet avant-garde narrative.	19 records

Propagation chain

Both datasets reveal the same downstream structure. An authority record at the Library of Congress that lacks a nationality qualifier does not simply remain neutral — the absence propagates as if it were a

positive assertion. The chain moves from authority record → aggregator (WorldCat) → national catalogue (Trove) → institutional catalogue (museum) → LLM training data → public-facing AI systems. At each stage the error becomes more entrenched and harder to correct upstream.

The tool has two components operating in parallel: a backend detection engine and a frontend exhibition system. Both are built to work offline or with live cloud LLMs depending on the exhibition context.

Multi-LLM misattribution detection

The tool queries three LLM systems with an identical prompt about each artist and compares the responses against the researcher-built Record of Truth, demonstrating that misattribution is baked into multiple AI systems through their training data.

System	Type	Finding
Claude (Anthropic)	Cloud LLM	Describes Malevich as "Russian avant-garde artist"; Parajanov as "Soviet film director"
GPT-4o (OpenAI)	Cloud LLM	Same misattribution — independently confirms "Russian avant-garde", "Soviet film director"
LLAMA 3 (Ollama)	Local / offline LLM	Same misattribution — confirms the error is model-independent, not specific to one system

All three systems — cloud and local, from different organisations — produce the same misattribution independently. This demonstrates that the error is not the failure of a single system but a structural condition of the training data.

Visitor kiosk

A touch screen at the entrance allows visitors to select an artist and tap "Ask the archive". The screen splits into four columns: one for each of the three LLM systems, and a fourth showing the researcher's Record of Truth. Each LLM response appears character by character. A bar at the bottom fills to show what proportion of AI systems produced the misattribution. No prior knowledge of the artists is required.

API endpoints

Endpoint	Function
/api/compare?artist=X	Returns all three LLM responses + Record of Truth for artist X
/api/query?artist=X	Single Claude query with misattribution assessment
/api/detect?url=IMG	Vision query — sends artwork image to LLM, returns assessment
/api/images?artist=X	Image manifest showing which artworks are ready for vision queries
/api/stats	Dataset summary statistics

The exhibition display runs as a continuous loop across 12 LED screens assembled together. Each screen runs a fullscreen browser window showing its section of a shared canvas. Together they compose one large image. The default grid is 4 columns × 3 rows; this can be reconfigured to 3×4, 6×2, or a linear 12×1 arrangement.

Visual language

The exhibition uses no text and no colour coding tied to national identity. The only visual language is **distortion** (misattribution) and **clarity** (truth). The palette is neutral throughout — black, white, and grey. Fragmentation, pixelation, glitch, and noise show the corrupted state. A whole, sharp, luminous image shows the corrected state.

Eight phases — continuous loop

01 PRISTINE The artwork appears clean and whole — a quiet white glow. The image as it was made, before the archive touched it.	02 LABEL APPLIED A dark rectangular mark descends onto the image — no text, no colour, just weight and opacity. The wrong classification settles.
03 CORRUPTION The image fractures. Pixels break into blocks, glitch bars tear across, shards drift away. The image as the archive made it.	04 PROPAGATION Distorted copies appear at the edges of the 12 screens. Archive nodes activate — library, museum, AI — the error flows downstream.
05 MACHINE GAZE Three scan bars of different intensity sweep across the corrupted image — cloud LLM, local LLM, library system. All confirm the wrong answer.	06 INTERVENTION A single white vertical beam cuts through. The distortion dissolves around it. The researcher's correction enters.
07 TRUTH The image returns — whole, sharp, luminous. Smaller correct copies orbit the centre, connected by fine lines. The work restored.	08 FADE The screens go dark. The cycle resets with the next artist. The loop continues — as the error does, until corrected.

Screen synchronisation

When all 12 screens run from a single exhibition computer, phase timing is shared via the browser's `localStorage` — screen 1 writes the clock, screens 2–12 read it. All 12 advance through the phases in lockstep. For multi-machine deployments, a server sync endpoint provides the shared clock across a local network.

Artwork images

The visualization displays actual artwork images when available, loaded from an image directory mapped to dataset record IDs (MAL-001, EXT-001, KAK-001, PAG-001, etc.). 78 records have direct image URLs; 16 of 19 Kakabadze records at Yale are open-access and ready for LLM vision queries without additional licensing. When images are provided via Google Drive, the image pipeline downloads and maps them automatically.

```
Teona-Yamanidze/

■■■ index.html           ← exhibition screen (one per display device)
■■■ exhibition.html      ← curator controller + dataset analysis
■■■ kiosk.html           ← visitor touch screen (3 LLMs vs. Record of Truth)
■■■ js/
■   ■■■ data.js          ← artist identities, phase definitions
■   ■■■ sketch.js        ← p5.js 8-phase visual narrative
■   ■■■ sync.js          ← 12-screen phase synchronisation
■■■ server/
■   ■■■ server.py        ← HTTP server + all API routes
■   ■■■ multi_llm.py     ← Claude + GPT-4o + LLAMA comparison
■   ■■■ image_pipeline.py ← Drive folder → local image index
■   ■■■ claude_detector.py ← standalone CLI misattribution detector
■■■ artworks_data.json   ← 354 artwork records (processed from Excel)
■■■ parajanov_data.json  ← 305 bibliographic records
```

Dependencies

Library	Purpose
p5.js 1.9	Exhibition visualization — browser-based, no build step
anthropic ≥0.40	Claude API — LLM queries and vision analysis
openai ≥1.30	GPT-4o — optional second LLM column
ollama (local)	LLAMA 3 — offline, no API key or internet required
openpyxl	Reading the Excel datasets
gdown	Google Drive image folder download

To run

```
pip install anthropic openpyxl

export ANTHROPIC_API_KEY=sk-ant-...

python3 server/server.py
```

```
# Open in browser:

# Exhibition → http://localhost:8000/index.html?screen=1&cols;=4&rows;=3

# Kiosk      → http://localhost:8000/kiosk.html

# Controller → http://localhost:8000/exhibition.html
```

Adding artwork images from Google Drive

```
pip install gdown

python3 server/image_pipeline.py --folder-id YOUR_DRIVE_FOLDER_ID

# Image filenames should include artwork IDs, e.g.:

# MAL-001_suprematist_teapot.jpg

# EXT-001_theatrical_composition.png

# PAG-001_grande_ville.jpg
```

The tool is designed to test how provenance can work as infrastructure within a museum exhibition context. It demonstrates three interlinked conditions identified through the dataset research:

1. Metadata errors are replicated and entrenched.

When the Library of Congress creates or maintains an authority record without a nationality qualifier, that absence propagates through every system that inherits from it. WorldCat, Trove, institutional catalogues — all downstream systems receive and replicate the gap as if it were a positive fact.

2. These entrenchments are replicated in LLMs.

Large language models trained on web-scale data absorb the cataloguing errors of the institutions that produced that data. The tool demonstrates this empirically: Claude, GPT-4o, and LLAMA all independently describe Malevich as a "Russian avant-garde artist" and Parajanov as a "Soviet film director" — because those are the labels that dominate the digital record.

3. Correction is not obvious, and the ground truth struggles to go back upstream.

The researcher has built a parallel Record of Truth — three additional columns for each of the 659 records showing what the nationality label, name form, and subject headings should have been. The exhibition makes visible both the error and the correction, and the difficulty of the distance between them.

The exhibition is therefore not only an artwork. It is a demonstration of infrastructure — of how digital systems shape what can be discovered, legitimised, and remembered within cultural memory.